Rehab: A Camera-Based 3D Rehabilitation Trainer for Arm Mobility

Can a webcam score rehab exercises? A first test with three users

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Abstract

Rehab needs many repetitions, but at home nobody checks if the movement is correct or measures progress. I built Rehab, a web app that tracks the hand with a webcam (MediaPipe), shows a 3D hand that copies the user, and gives an accuracy score for each exercise. Three users made 32 analyses; average accuracy was 77–96%, and in one user supination accuracy rose from 58% to 87% between the first and the second try. A camera can give cheap, instant feedback for home practice, but my test is small and is not a medical result.

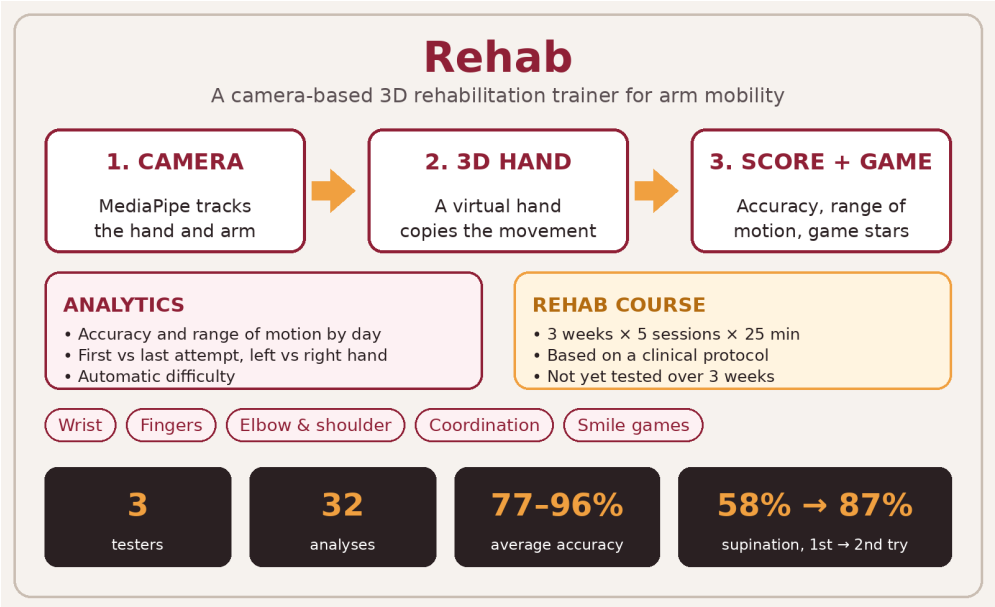


Figure 1: Rehab: overview of the system and the main results.

1 Introduction

After an injury, a doctor gives exercises that must be repeated many times a week. At home they are boring, nobody sees if the form is right, and progress is a feeling, not a number.

Many apps show videos of exercises, but they do not check the movement. I built a web app that watches the user with a normal webcam and scores each exercise.

My contributions are: (1) a web trainer with exercises for wrist, fingers, elbow, shoulder, coordination and smile; (2) a 3D hand model and games that give feedback; (3) an analytics page and a rehabilitation course plan; (4) a first test with three users and 32 analyses.

2 Related work

MediaPipe Hands [2] finds 21 points of a hand in real time from one camera, and is part of the MediaPipe framework [1]. I use it to read the hand. Those works describe tracking; my work uses the points to score rehab exercises and to run games, and I report a first test with users.

3 What I built

The user stands in front of a webcam. MediaPipe Hands finds the hand points; I compute joint angles (pitch, yaw, roll for the wrist; finger bend) and show them on a 3D hand built with Three.js. For each exercise the app shows a demo, the user repeats it, the app gives an accuracy score and the range of motion (degrees for joints, per cent for fingers), and a short game follows. The app runs in the browser, for the left or the right hand, with calibration settings. I did not train a new model.

The app also has an analytics page. It shows accuracy by day, range of motion by day, range of motion before and after (first attempt against last attempt), the results for the right and the left hand, and the results for each mode. It can also choose the difficulty automatically. Next to the analytics there is a rehabilitation course: 3 weeks with 5 sessions of 25 minutes each, built after a clinical protocol, with a button to start it. The course plan is part of the app, but I have not tested it over 3 weeks.



Figure 2: How it works: webcam → hand tracking and 3D model → score and game.

4 How I measure

Three users (A, B, C) used the app in one session each (4 minutes or less). A used the left hand (3 analyses), B the left hand (11), C the right hand (18); 32 analyses in total. The app gives an accuracy score for each analysis. For exercises that a user repeated, the **baseline** is the first attempt, and I compare it with the last attempt. All numbers come from the reports that the app saves.

5 Results

Average accuracy over all analyses was 96% for A, 81% for B and 77% for C. Finger exercises (fist and open, finger wave, gestures) scored 93–100%; rotation and combined movements scored lower (55–66% on the first try). Only two exercises were repeated by one user (C). Table 1 compares the first and last attempt.

Supination accuracy rose from 58% to 87% in the second attempt, but its range of motion changed from 105 to 102 degrees. The 3-joint exercise got worse, from 66% to 62% (three attempts; the lowest was 40%). Numbers move between attempts, and each exercise was done only a few times.

Attempt (user C)	Supination ↑	3-joint exercise ↑
First attempt (baseline)	58%	**66%**
Last attempt	**87%**	62%

Table 1: Accuracy of the first and last attempt, for the two exercises that user C repeated (2 and 3 attempts). ↑ = higher is better.

6 Limitations

Only three users took part, and they made one session each, so the result can be wrong for other people. Most exercises were done once, so I cannot show improvement over time. The 3-week course was not tested: nobody used it for 3 weeks, so I have no result for it. The users were healthy; I did not test with patients. The camera measures movement only, not grip force. Rehab is not a medical device and does not replace a doctor.

7 Ethics

Testers are named A, B and C; I publish no names, faces or videos. The testers agreed to take part. I used MediaPipe (Apache 2.0) and Three.js (MIT). I used an AI agent to write code.

8 Conclusion

I built a web trainer that scores rehab exercises from a webcam. In a first test with three users, average accuracy was 77–96%, and supination accuracy rose from 58% to 87% from the first to the second try, but the 3-joint exercise did not improve. Next I want to test with patients and a doctor, repeat exercises over several days, and improve the hardest movements. The 3-week course must be tested with real users.

Acknowledgments

I thank my mentors at ISSAI and the FRP 2026 program. Project page: [https://adiatursyn77-bit.github.io/rehab-page/].

References

[1] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, et al. MediaPipe: A framework for building perception pipelines, 2019.

[2] Fan Zhang, Valentin Bazarevsky, Andrey Vakunov, Andrei Tkachenka, George Sung, Chuo-Ling Chang, and Matthias Grundmann. MediaPipe Hands: On-device real-time hand tracking, 2020.